

AGGREGATE TRENDS

PRESENTATION 2

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Thursday, October 23rd



Redirection Analysis

Understanding how and why calls are redirected for the top called numbers.

1. Define Webex redirection fields
2. Example call path
3. Redirect vs. original reasons (top numbers)
4. Granular front desk analysis
5. Recommendations
6. Next Steps

Each call can generate multiple legs, redirections help explain how calls move through the system.

Call Start

- **Calling Number** → number that initiates the call
- **Direction = TERMINATING** → indicates the call is inbound
- **Correlation ID** → groups all call legs of the same session

Call Movement

- **Called number** → initial recipient
- **Redirecting Number** → who last redirected the call
- **Original Reason** → first trigger for a redirection
- **Redirect Reason** → why subsequent redirection occurs
- **Related Reason** → the trigger that led to a change in call presence

Call End

- **User Type** → identifies the endpoint (User, Voicemail, etc.)
- **Answered** → whether the call was answered or not
- **Report Time** → when the call ended

Call Path for Correlation ID '9108b891-9c8f-4056-a98c-dea3d0a121a8' (March 2025).

*Client initially calls the
Ombudsman Nursing Home
line.*

Called number =
13122296079, Nursing
Home



Direction = Terminating



Original reason = No
Answer

Redirecting number =
13123478300



Redirect reason = No
Answer



User type = Virtual line



Related reason = Call
Forward No Answer

Model = N/A



User = Voice Mail Retrieval



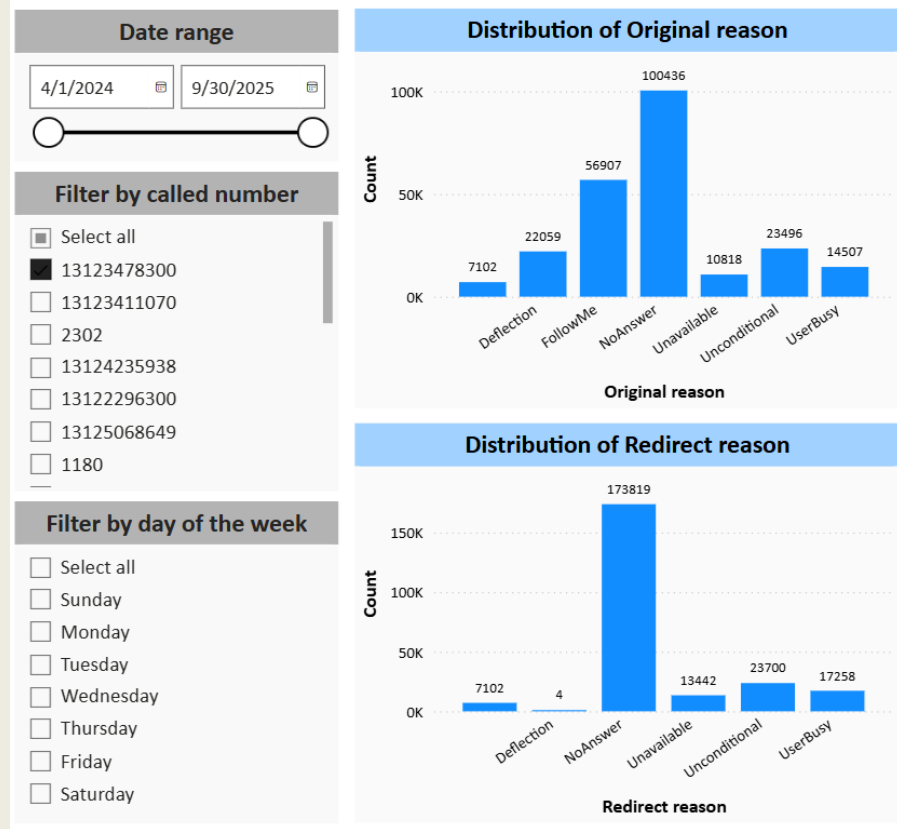
Call outcome = SUCCESS



Answered = TRUE

*The call was ultimately “answered”, but not by a person. It was **answered after being redirected to voicemail**.*

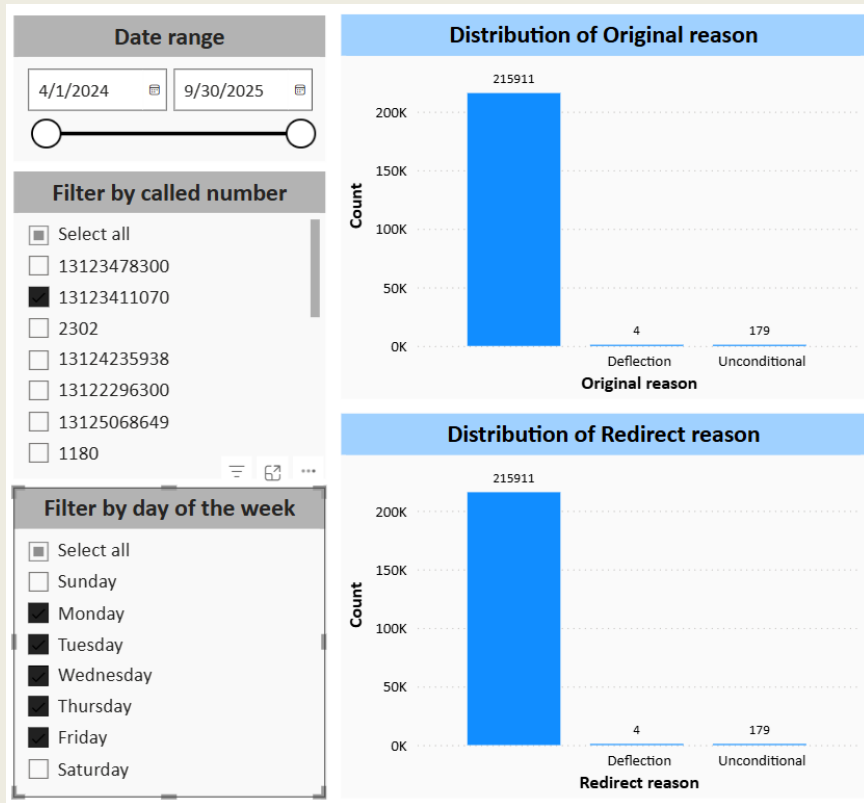
Distributions of original/redirect reasons for internal voicemail (13123478300) reflect its role as the ending leg of failed calls.



What We Found:

- Original reason is **predominantly NoAnswer** for internal voicemail, reflecting how it is the **outcome of failed calls**.
- Deflection** is an **extremely rare redirect reason** for 8300 → 8300 is an endpoint
- A call ending in 8300 could be used as an **indication of failure** → possibly helpful in calculating success rate metrics

Most phone numbers do not significantly differ in their original/redirect distributions between weekday calls and weekend calls.



What We Found:

- For many of the phone numbers that I created as a filter, there is no major difference between their reason distributions on weekdays versus weekends
- Specifying hours of the day to look at will likely be more fruitful than trying to analyze the data at the level of days of the week

Date range

4/1/2024

9/30/2025

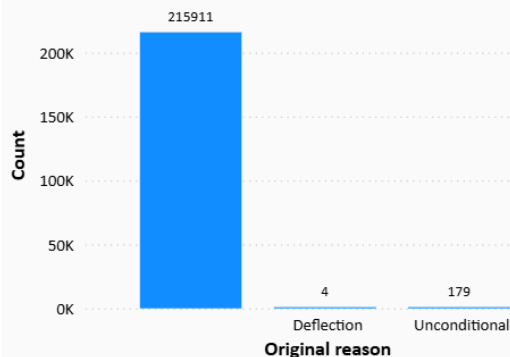
Filter by called number

- ☒ Select all
- ☐ 13123478300
- ☒ 13123411070
- ☐ 2302
- ☐ 13124235938
- ☐ 13122296300
- ☐ 13125068649
- ☐ 1180

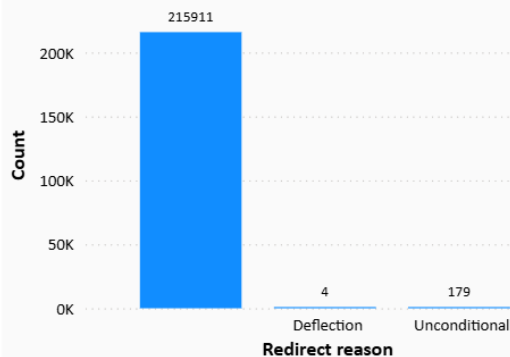
Filter by day of the week

- ☒ Select all
- ☐ Sunday
- ☒ Monday
- ☒ Tuesday
- ☒ Wednesday
- ☒ Thursday
- ☒ Friday
- ☐ Saturday

Distribution of Original reason



Distribution of Redirect reason



Date range

4/1/2024

9/30/2025

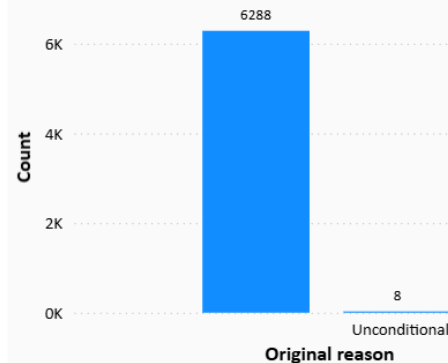
Filter by called number

- ☒ Select all
- ☐ 13123478300
- ☒ 13123411070
- ☐ 2302
- ☐ 13124235938
- ☐ 13122296300
- ☐ 13125068649
- ☐ 1180

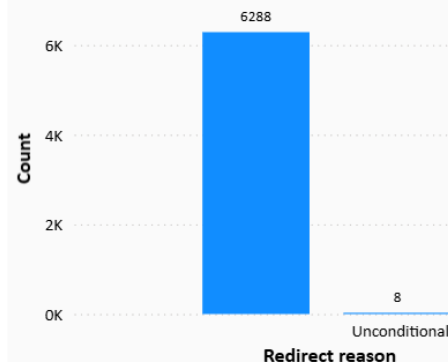
Filter by day of the week

- ☒ Select all
- ☒ Sunday
- ☐ Monday
- ☐ Tuesday
- ☐ Wednesday
- ☐ Thursday
- ☐ Friday
- ☒ Saturday

Distribution of Original reason

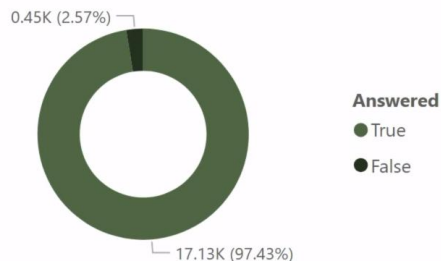


Distribution of Redirect reason

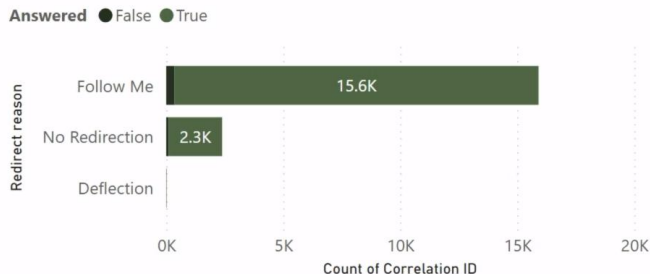


Front desk calls are being answered efficiently, with most actively managed through automatic routing.

Answered vs Not Answered (Front Desk)



Redirect reason for Front Desk Calls

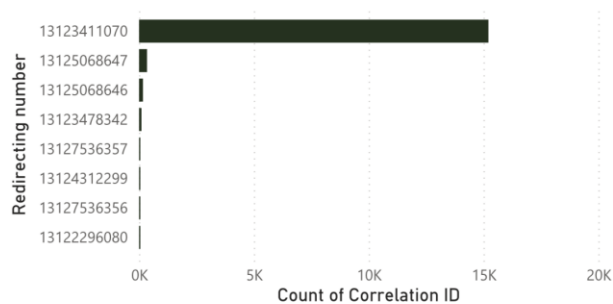


What We Found:

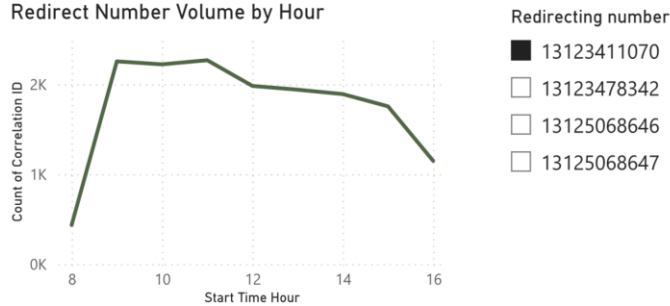
- ~ **97% of front desk calls are answered** showing strong coverage overall
- **The dominant redirect reason is “Follow Me”**, meaning the calls are automatically routed through Webex’s sequential forwarding logic until answered
- **Most redirected calls still reach a person**, showing that automated routing is working effectively to prevent missed calls for this line

Most “Follow Me” redirections from the front desk route through the main number, peaking mid-day and tapering after 4 PM.

Count of Correlation ID by Redirecting number



Redirect Number Volume by Hour



What We Found:

- 15.6k redirected calls route through the **main number (13123411070)**; it seems to act as the system hub
- Specialized menus handle smaller volumes, **Spanish (13125068647)** ~ 347 calls & **English (13125068646)** ~ 170 calls
- Redirect call volumes for the main line from the front desk peak between **9 AM and 11 AM**, aligning with business hours

Recommendations

1

Reallocate agent availability to areas that often result in “No Answer”. This would reduce the rate of calls ending in voicemail. However, practically, this might be infeasible due to costs.



2

Add bilingual auto-menu routing at the front-desk number to reduce unnecessary “Follow Me” transfers. A brief bilingual menu at the front desk entry point would let callers self-select their language immediately, decreasing redirections from this line.



Questions & Next Steps

Open Questions

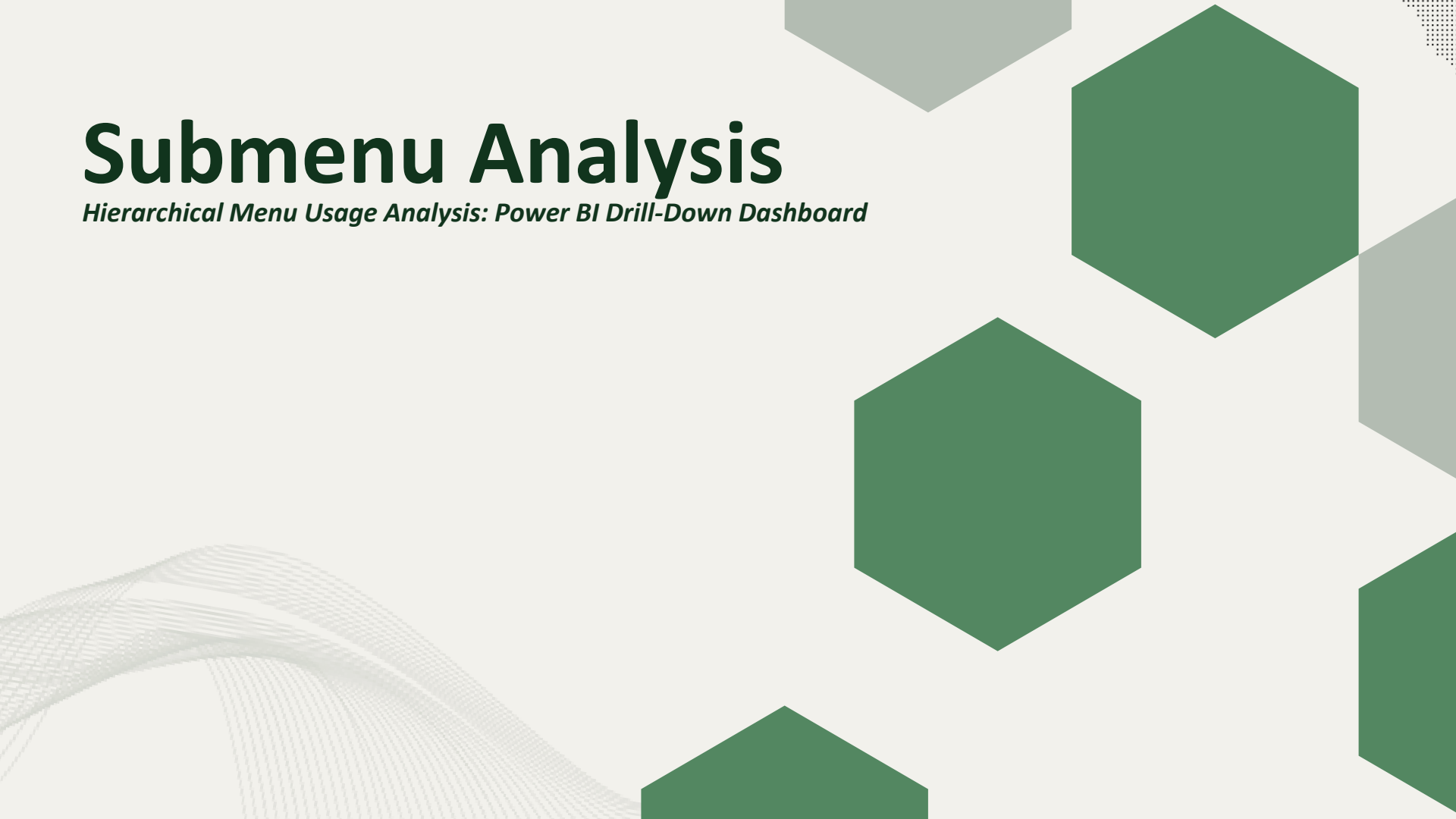
- What is the Follow Me routing logic in Webex? Can the configuration be accessed directly or can we only infer it through All Call data?
- What are the use cases for callers dialing the front desk directly instead of the main line?

Next Steps

1. **Analyze Patterns Behind Voicemail.** Identify trends in **Reason**, **User Type** and **Client Type** for calls that end in extension 8300 (internal voicemail) to isolate whether missed calls are from staffing gaps or menu inefficiencies.
2. **Expand Use of “Related Reason” Field.** Investigate distinctions between **Deflection** and **Consultative Transfer** to better understand how and why front-desk calls move between lines.

Submenu Analysis

Hierarchical Menu Usage Analysis: Power BI Drill-Down Dashboard



Defining the Hierarchy and Variables Used

EP (Entry Point)

Main menu where a call begins (e.g. Legal Menu, Housing Menu).

Flow

A submenu or routing option under an EP.

Activity

The final action or step reached within a call (e.g. Language Selection, Queue).

Contact Session ID

Unique identifier for each distinct call. Counted each time the ID is within a menu.

Submenu Count

Number of submenus navigated during a single call.

Unknown Values

Placeholder for missing data. Ensures complete hierarchy display.

Preparing the Data for Drill-Down Analysis

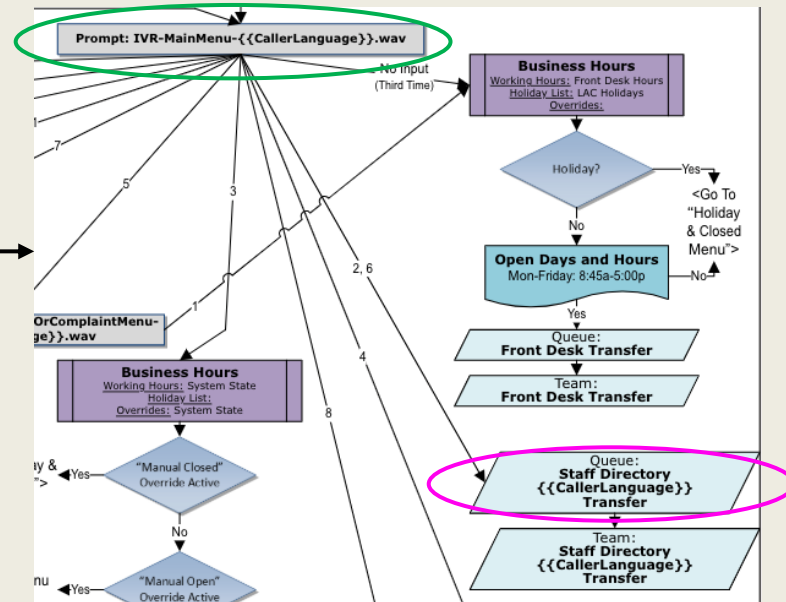
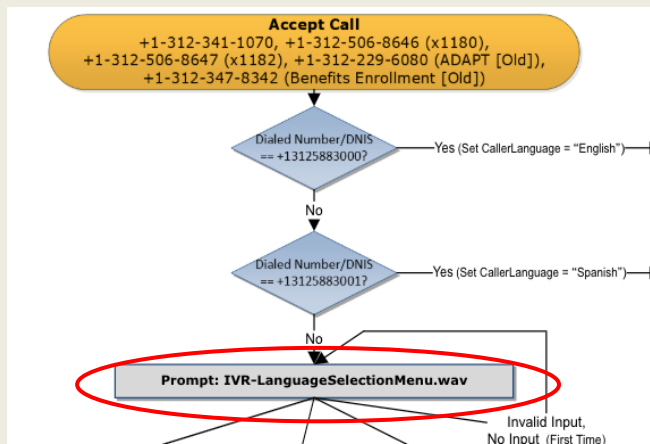
Contact Session ID	EP Name	Flow Name	Activity Name	Activity Start Timestamp	hour
001a3748-8d50-4550-8461-33547983de0	Main Number Telephone EP			1/14/25 12:39	12
001a3748-8d50-4550-8461-33547983de0		LACMain		1/14/25 12:39	12
001a3748-8d50-4550-8461-33547983de0	Main Number Telephone EP		LanguageSelectionMen	1/14/25 12:39	12
001a3748-8d50-4550-8461-33547983de0	Main Number Telephone EP	LACMain		1/14/25 12:39	12
001a3748-8d50-4550-8461-33547983de0	Main Number Telephone EP	MainMenu		1/14/25 12:39	12
001a3748-8d50-4550-8461-33547983de0		PreLegalMenuSeniorsMenu		1/14/25 12:40	12
001a3748-8d50-4550-8461-33547983de0	Pre-Legal Menu Seniors Menu	SeniorsMenu		1/14/25 12:40	12
001a3748-8d50-4550-8461-33547983de0		LegalMenu		1/14/25 12:40	12
001a3748-8d50-4550-8461-33547983de0	Legal Menu Telephone EP	LegalMenu1		1/14/25 12:40	12
001a3748-8d50-4550-8461-33547983de0	Legal Menu Telephone EP	LegalMenu2		1/14/25 12:40	12
001a3748-8d50-4550-8461-33547983de0		LegalEmploymentMenu		1/14/25 12:43	12
001a3748-8d50-4550-8461-33547983de0	Legal Employment Menu Telephone EP	EmploymentMenu		1/14/25 12:43	12
001a3748-8d50-4550-8461-33547983de0		Queues		1/14/25 12:43	12
001a3748-8d50-4550-8461-33547983de0		ClosedQueueMenu		1/14/25 12:43	12
001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephone EP	ClosedQueueMenu		1/14/25 12:43	12
001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephone EP			1/14/25 12:44	12
001a3748-8d50-4550-8461-33547983de0	Closed Queue Menu Telephone EP			1/14/25 12:44	12
00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephone EP			1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		LACMain		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephone EP		LanguageSelectionMen	1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Main Number Telephone EP	LACMain		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		MainMenu		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		PreLegalMenuSeniorsMenu		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Pre-Legal Menu Seniors Menu Telephone EP	SeniorsMenu		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		LegalMenu		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Menu Telephone EP	LegalMenu1		1/14/25 14:50	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Menu Telephone EP	LegalMenu2		1/14/25 14:51	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		LegalFamilyMenu		1/14/25 14:52	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Legal Family Menu Telephone EP	FamilyMenu		1/14/25 14:52	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		Queues		1/14/25 14:52	14
00229391-8614-4eb1-b1e0-e0a78076e0e4		ClosedQueueMenu		1/14/25 14:52	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephone EP	ClosedQueueMenu		1/14/25 14:52	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephone EP			1/14/25 14:53	14
00229391-8614-4eb1-b1e0-e0a78076e0e4	Closed Queue Menu Telephone EP	ClosedQueueMenu		1/14/25 14:53	14
0020c673-c0c5-42f4-3046-e0b9ac7080b	Main Number Telephone EP			1/13/25 10:54	10

contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	hour	submenu_count
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	Unknown Flow	Unknown Activity	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	LanguageSelection	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	MainMenu	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	StaffDirectoryEnglis	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
00002422-f51f-458b-8206-cfa5a3f36f69	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-13T12:52:00Z	12	10
00014a58-afce-4b2d-a529-d5e2dc304d	Main Number Telephone EP	Unknown Flow	Unknown Activity	2025-02-28T08:33:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Main Number Telephone EP	LACMain	Unknown Activity	2025-02-28T08:33:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Main Number Telephone EP	LACMain	LanguageSelection	2025-02-28T08:33:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Main Number Telephone EP	LACMain	Unknown Activity	2025-02-28T08:33:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Pre-Legal Menu Seniors Menu Telephone EP	PreLegalMenuSeniorsMenu	SeniorsMenu	2025-02-28T08:34:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Pre-Legal Menu Seniors Menu Telephone EP	LegalMenu	Unknown Activity	2025-02-28T08:34:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Legal Menu Telephone EP	LegalMenu	LegalMenu1	2025-02-28T08:34:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Legal Menu Telephone EP	LegalMenu	LegalMenu2	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Legal Menu Telephone EP	LegalFamilyMenu	Unknown Activity	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Legal Family Menu Telephone EP	LegalFamilyMenu	FamilyMenu	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Queues	Queues	Unknown Activity	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Closed Queue Menu Telephone EP	ClosedQueueMenu	Unknown Activity	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Closed Queue Menu Telephone EP	ClosedQueueMenu	ClosedQueueMenu	2025-02-28T08:35:00Z	8	15
00014a58-afce-4b2d-a529-d5e2dc304d	Closed Queue Menu Telephone EP	ClosedQueueMenu	Unknown Activity	2025-02-28T08:35:00Z	8	15
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	Unknown Flow	Unknown Activity	2024-05-29T10:55:00Z	10	5
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	LACMain	Unknown Activity	2024-05-29T10:55:00Z	10	5
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	LACMain	LanguageSelection	2024-05-29T10:55:00Z	10	5
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	LACMain	Unknown Activity	2024-05-29T10:55:00Z	10	5
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	LACMain	MainMenu	2024-05-29T10:55:00Z	10	5
00023e47-6722-4e2f-b2e2-855911d085f4	Main Number Telephone EP	LACMain	Unknown Activity	2024-05-29T10:56:00Z	10	5
0003369c-5377-4d88-9c56-779a75d26e22	Main Number Telephone EP	Unknown Flow	Unknown Activity	2025-03-12T10:49:00Z	10	10
0003369c-5377-4d88-9c56-779a75d26e22	Main Number Telephone EP	LACMain	Unknown Activity	2025-03-12T10:49:00Z	10	10

What We Did:

- Standardized/combined CAR dataset
- Created submenu_count variable
- Replaced NAs with “Unknown” placeholders
- Converted ep_name, flow_name, and activity_name to text
- Filled missing EPs and Flows by filling from the row above and grouping by contact_session_id

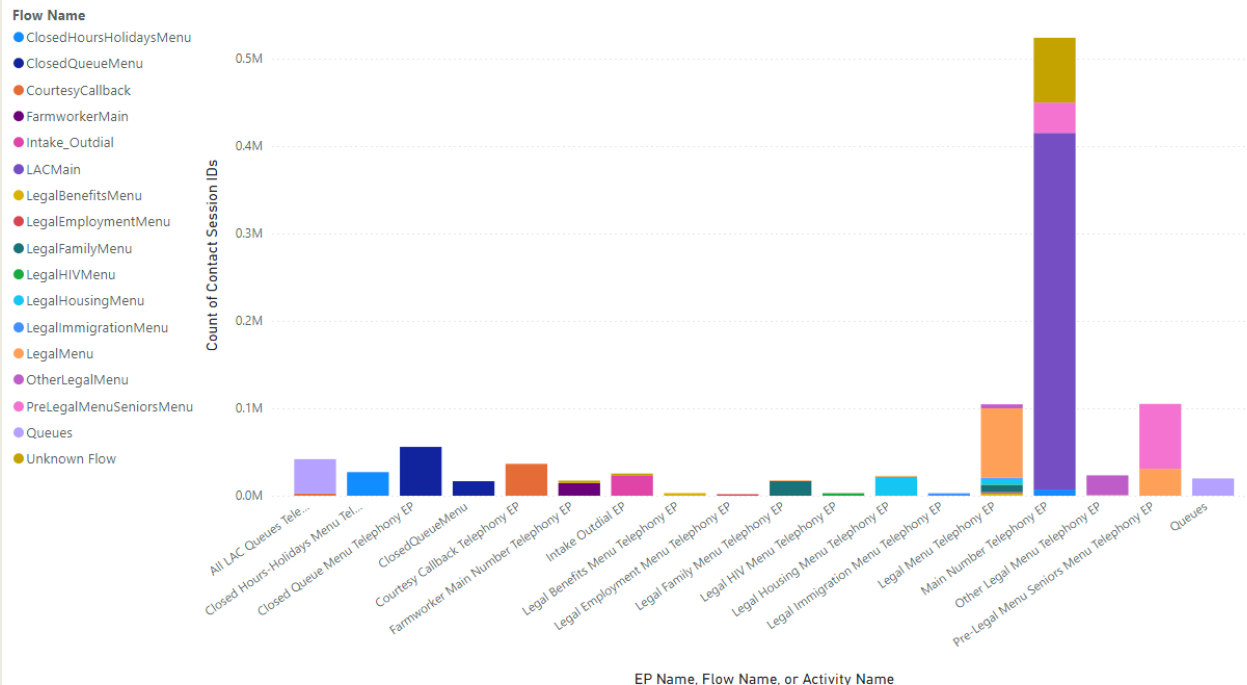
Call Flow Example



	contact_session_id	ep_name	flow_name	activity_name	activity_start_timestamp	hour	submenu_count
1	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	Unknown Flow	Unknown Activity	2025-03-13T12:51:00Z	12	10
2	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
3	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	LanguageSelectionMenu	2025-03-13T12:51:00Z	12	10
4	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
5	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	Unknown Activity	2025-03-13T12:51:00Z	12	10
6	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	MainMenu	2025-03-13T12:51:00Z	12	10
7	00002422-f51f-458b-82d6-cfa5a3f36fd9	Main Number Telephony EP	LACMain	StaffDirectoryEnglish	2025-03-13T12:51:00Z	12	10

Baseline Distribution of Contact Session IDs by EP Name, Colored by Flow Name

Count of Contact Session IDs by EP Name, Flow Name, and Activity Name



What We Found:

- The **Main EP** is visited most frequently; indicates we should prioritize simplifying this EP since it takes extra time to read out all these options
- Under the Main Menu, there are **12 Activity Names**, which takes **58 seconds** to list; separating these activities into 2 groups could improve efficiency
- Benefits, Employment, HIV, Immigration EPs** are rarely used; could combine these to simplify the system then have them be just flows instead of their own EPs

Recommendations

1

Simplify high-complexity menus



2

Develop Menu Efficiency Metrics



3

Reduce Queue-Related Bottlenecks



Next Steps & Further Questions

- Add interactive filters and KPIs: Include slicers for date, day type, and hours; add cards for total calls and average submenus per EP
 - Develop efficiency indicators: Calculate average submenu depth per EP or Flow to highlight areas with the most caller confusion or unnecessary steps
 - Group related menus: Merge similar Eps (e.g. Benefits, Employment, HIV, Immigration) for clearer visualization and comparison
 - Group related activities: there are 12 options inside of Main Menu → splitting into two groups could improve efficiency
1. Does the EP → Flow → Activity hierarchy in our cleaned data match LAC's structure?
 2. Can Cynthia clarify what the Flow Name means?
 3. Would LAC be open to restructuring or merging high-friction menus based on our findings?
 4. Which metrics are most important to LAC between minimizing submenu steps, reducing queue time, or ensuring correct routing on the first try?

Appendix

** Includes additional details and visuals from our analysis that were not included in the live presentation due to time constraints.*



Data Cleaning (Python)

The following data cleaning was completed in **Python** for the **All Calls Dataset**. The new data from *September 2025* was also added to the combined code csv. Code [here](#).

Converting date to datetime format

```
In [59]: df_main["Start time"] = pd.to_datetime(df_main["Start time"], utc=True)
```

```
In [60]: df_main.to_csv("combined_calls.csv", index=False)
```

```
In [61]: df_main["Start time"] = df_main["Start time"].dt.tz_convert("America/Chicago").dt.tz_localize(None)
```

```
In [62]: df_main["Start time"].head()
```

```
Out[62]: 0    2024-04-30 18:58:53.988
1    2024-04-30 18:56:37.386
2    2024-04-30 18:54:59.099
3    2024-04-30 18:54:59.099
4    2024-04-30 18:54:52.336
Name: Start time, dtype: datetime64[ns]
```

```
In [63]: # Extract hour of day
df_main["Start Time Hour"] = df_main["Start time"].dt.hour
print(df_main[["Start time", "Start Time Hour"]].head())
```

	Start time	Start Time Hour
0	2024-04-30 18:58:53.988	18
1	2024-04-30 18:56:37.386	18
2	2024-04-30 18:54:59.099	18
3	2024-04-30 18:54:59.099	18
4	2024-04-30 18:54:52.336	18

```
In [64]: df_main.to_csv("combined_calls_with_start_time_hour.csv", index=False)
```